MongoDB Lab (M23BCS408B)

Lab 1:

Aim: Introduction to NoSQL and MongoDB 1a. Install MongoDB on your system (or use a pre-configured instance). Connect to the database using the MongoDB shell. Verify your connection by listing all the databases.

open command prompt

mongod --version

db version v7.0.8

mongosh

Using MongoDB: 8.0.4

Using Mongosh: 2.2.12

test> **show dbs**

admin 40.00 KiB

config 60.00 KiB

local 40.00 KiB

test 112.00 KiB

1b. Create a new database named 'my_database'. Switch to this database and create a collection named 'students'. Insert two sample documents into the 'students' collection, each representing a student with the following fields: 'name', 'age', 'city'.

test> use mydatabase

switched to db mydatabase

mydatabase>

Create the students Collection:

To create the students collection, use the createCollection() method.

This step is optional because MongoDB will automatically create the collection when you insert data into it, but you can explicitly create it if needed:

mydatabase> db.createCollection("students")

>use test switched to db

{ "ok" : 1 }

mydatabase> >show collections

students

Insert operations

Insert a Single Document into students

mydatabase> db.students.insertOne({ name: "abhi",age: "20", city: "mysore"})

true:ok;

{ "_id" : ObjectId("5d7d3f28315728b4998f522f"),

Find All Documents

To retrieve all documents from the students collection:

mydatabase> db.students.find()

{ "_id" : ObjectId("5d7d3f28315728b4998f522f"), name: "abhi",age: "20", city: "mysore" }

2a. Add a new field 'email' to the existing student documents in the 'students' collection. Update the documents to include email addresses for each student.

my_database> use school

switched to db school

school> show collections

```
school>
db.students.insertMany([{name:"ram",age:"24",city:"mysore"},{name:"shyam",age:"32",city:"mandya"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67b997ee04cb32daa9117b7f'),
    '1': ObjectId('67b997ee04cb32daa9117b80')
  }
}
school> db.students.updateMany({},{$set:{email:""}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
school> db.students.updateOne({name:"ram"},{$set:{email:"ram544@gmail.com"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
school> db.students.updateOne({name:"shyam"},{$set:{email:"shyam544@gmail.com"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
school> db.students.find()
[
  {
    _id: ObjectId('67b997ee04cb32daa9117b7f'),
    name: 'ram',
    age: '24',
    city: 'mysore',
    email: 'ram544@gmail.com'
  },
  {
    _id: ObjectId('67b997ee04cb32daa9117b80'),
    name: 'shyam',
    age: '32',
    city: 'mandya',
    email: 'shyam544@gmail.com'
  }
]
```

2b: Write a MongoDB query to retrieve all student documents with an age greater than 20. Additionally, write another query to retrieve students whose city is 'Bengaluru'.Master CRUD operations (Create, Read, Update,Delete) on MongoDB document

```
school> db.students.find({age: {$gt:"20"} })
[
  {
    _id: ObjectId('67b997ee04cb32daa9117b7f'),
    name: 'ram',
    age: '24',
    city: 'mysore',
    email: 'ram544@gmail.com'
  },
  {
    _id: ObjectId('67b997ee04cb32daa9117b80'),
    name: 'shyam',
    age: '32',
    city: 'mandya',
    email: 'shyam544@gmail.com'
  }
]
school> db.students.updateOne({name:"ram"},{$set:{city:"banglore"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
school> db.students.find({city:"banglore"})
[
  {
    _id: ObjectId('67b997ee04cb32daa9117b7f'),
    name: 'ram',
    age: '24',
    city: 'banglore',
    email: 'ram544@gmail.com'
  }
]
```

3a) Use the '$regex' operator to retrieve all student documents whose name starts with the letter 'A'. Then, retrieve students whose city contains the word "Delhi".

```
db.Students.insertMany([{name:"amar",age:"24",city:"banglore"},{name:"prathik",age:"25", city:"delhi"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67c67f3546adc10b04117b7b'),
    '1': ObjectId('67c67f3546adc10b04117b7c')
  }
}
```

```
test> db.Students.find({name:{$regex:"^A", $options:"i"}})
[
  {
    _id: ObjectId('67c67f3546adc10b04117b7b'),
    name: 'amar',
    age: '24',
    city: 'banglore'
  }
]
test> db.Students.find({city:{$regex:"delhi", $options:"i"}})
[
  {
    _id: ObjectId('67c67f3546adc10b04117b7c'),
    name: 'prathik',
    age: '25',
    city: 'delhi'
  }
]
```

3b) Create a new collection named 'books' and populate it with sample documents representing different books, each containing fields like 'title', 'author', and 'genre'. Write a query to retrieve all books that have the genre 'Science Fiction'

```
test> db.books.insertMany([{title:"dum",author:"Frank Herbert", genre:"Science
Fiction"},{tile:"Founda",author:"Isaac Asimov", genre:"Science Fiction"},{title:"1984",author:"George
Orwell", genre:"Dystopian"},{title:"Brave New World ", author:"Aldas Huxley",
genre:"Dystopian"},{title:"Neuromancer", author:"William Gibson", genre:"cyberpunk"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67c683d246adc10b04117b7d'),
    '1': ObjectId('67c683d246adc10b04117b7e'),
    '2': ObjectId('67c683d246adc10b04117b7f'),
    '3': ObjectId('67c683d246adc10b04117b80'),
    '4': ObjectId('67c683d246adc10b04117b81')
  }
}
test> db.books.find({genre:"Science Fiction"})
[
  {
    _id: ObjectId('67c683d246adc10b04117b7d'),
    title: 'dum',
    author: 'Frank Herbert',
    genre: 'Science Fiction'
  },
  {
    _id: ObjectId('67c683d246adc10b04117b7e'),
    tile: 'Founda',
    author: 'Isaac Asimov',
    genre: 'Science Fiction'
  }
```

```
]
test> db.books.find({genre:{$regex:"Science Fiction", $options:"i"}})
[
  {
    _id: ObjectId('67c683d246adc10b04117b7d'),
    title: 'dum',
    author: 'Frank Herbert',
    genre: 'Science Fiction'
  },
  {
    _id: ObjectId('67c683d246adc10b04117b7e'),
    tile: 'Founda',
    author: 'Isaac Asimov',
    genre: 'Science Fiction'
  }
]
```

Aim: Aggregation and Data Analysis

Aggregation in **MongoDB** is a powerful framework that allows developers to perform complex **data transformations**, computations and analysis on collections of documents. By utilizing the **aggregation** pipeline, users can efficiently **group**, **filter**, **sort**, **reshape**, and perform calculations on data to generate meaningful insights.

$count Calculates the quantity of documents in the given group.
$max Displays the maximum value of a document's field in the collection.
$min Displays the minimum value of a document's field in the collection.
$avg Displays the average value of a document's field in the collection.
$sum Sums up the specified values of all documents in the collection.
$push Adds extra values into the array of the resulting document.
$match stage – filters those documents we need to work with, those that fit our needs
$group stage – does the aggregation job
$sort stage – sorts the resulting documents the way we require (ascending or descending)

4a. Write an aggregation pipeline to calculate the average age of students in the 'students' collection.

```
db.Students.insertMany([
  { name: "Alice", age: 28, city: "New York" },
  { name: "Bob", age: 32, city: "Chicago" },
  { name: "Charlie", age: 45, city: "New York" },
  { name: "David", age: 39, city: "Los Angeles" },
  { name: "Eva", age: 27, city: "Chicago" },
  { name: "Frank", age: 51, city: "Los Angeles" }
])
db.Students.aggregate([
{
   $group: {
     _id: null,              // Group all documents together
     averageAge: { $avg: "$age" }  // Calculate the average of the 'age' field
    }}])
```

Output:

```
[ { _id: null, averageAge: 37 } ]
```

Example 2 when number is in string format like age:"25"

```
db.Students.aggregate([
  { $project: {
     age: { $toInt: "$age" }  // Convert the 'age' field from string to integer
   }
  }, {
    $group: {_id: null,              // Group all documents together
averageAge: { $avg: "$age" }  // Calculate the average of the 'age' field
    }
  }
])
```

Output:

```
[ { _id: null, averageAge: 37 } ]
```

4b. Use an aggregation pipeline to group students by city and count the number of students in each city. Then, sort the results in descending order based on the count.

```
db.Students.aggregate([
  {
    $group: {
      _id: "$city",              // Group by the 'city' field
      count: { $sum: 1 }          // Count the number of people in each city
    }
  },
  {
    $sort: { count: -1 }         // Sort the results in descending order based on 'count'
  }
])
```

Output:

```
[
  { _id: 'Chicago', count: 2 },
  { _id: 'Los Angeles', count: 2 },
  { _id: 'New York', count: 2 }
]
```

Aim: Indexing and Performance Optimization

5a. Create an index on the 'age' field in the 'students' collection. Execute a query to retrieve students with an age greater than 25 and compare its execution time before and after creating the index.

Indexing in MongoDB

- MongoDB uses indexing to make the query processing more efficient.
- If there is no indexing, then MongoDB must scan every document in the collection and retrieve only those documents that match the query.
- Indexes are special data structures that store some information related to the documents such that it becomes easy for MongoDB to find the right data file.
- The indexes are ordered by the value of the field specified in the index.

```
db.Students.insertMany([
  { name: "Alice", age: 20, city: "New York" },
  { name: "Bob", age: 32, city: "Chicago" },
  { name: "Charlie", age: 45, city: "New York" },
  { name: "David", age: 39, city: "Los Angeles" },
  { name: "Eva", age: 22, city: "Chicago" },
  { name: "Frank", age: 51, city: "Los Angeles" }
```

])
Finding students with an age greater than 25
mydatabase> db.Students.find({age: {$gt:25}})
mydatabase> db.Students.find({age: {$gt:20}}).explain("executionStats")
output: Before the index
stage: 'COLLSCAN',
filter: { age: { '$gt': 25 } },
executionStats: {
executionSuccess: true,
nReturned: 6,
executionTimeMillis: 5,
totalKeysExamined: 0,
totalDocsExamined: 6,
Create the index on the age **field:**
db.Students.createIndex({age:1})
age_1
To verify the created indexes, you can use the getIndexes **method.**
mydatabase> db.Students.getIndexes()
[
{ v: 2, key: { _id: 1 }, name: '_id_' },
{ v: 2, key: { age: 1 }, name: 'age_1' }
]
// The default index on the _id field (_id_).
// An index on the age field (age_1),

>db.Students.find({age: {$gt:25}}).explain("executionStats")
output: After the index
stage: 'IXSCAN'
executionStats: {
executionSuccess: true,
nReturned: 6,
executionTimeMillis: 14,
totalKeysExamined: 6,
totalDocsExamined: 4,
executionStages: {
isCached: false,
stage: 'FETCH'

5b. Create a compound index on 'city' and 'age' fields. Run a query to retrieve students from 'Mysuru' who are older than 20 and analyze the improvement in performance compared to a query without an index.

MongoDB Compound Indexes

- MongoDB Compound Indexes is an index on multiple fields in a MongoDB collection.
- It can be created to enhance the performance of queries that need to filter on multiple fields.
- A MongoDB compound index can cover queries that request only the indexed fields, improving efficiency by avoiding reading the actual documents.
- Compound Indexes do indexing on multiple fields of the document either in ascending or descending order i.e. it will sort the data of one field and then inside that it will sort the data of another field.

db.Students.find({ city: 'Mysuru', age: { $gt: 20 } })

Create the Compound Index:
db.students.createIndex({ city: 1, age: 1 })
city_1_age_1
db.Students.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { age: 1 }, name: 'age_1' },
  { v: 2, key: { city: 1, age: 1 }, name: 'city_1_age_1' }
]
db.Students.find({ city: 'Los Angeles', age: { $gt: 20 } }).explain("executionStats")
output:
executionStats: {
    executionSuccess: true,
    nReturned: 2,
    executionTimeMillis: 0,
    totalKeysExamined: 2,
    totalDocsExamined: 2,

To drop an index in MongoDB, you can use the dropIndex() **method. The syntax for this method is as follows:**
db.collection.dropIndex(index_name)
db.students.dropIndex("city_1_age_1")
db.Students.find({ city: 'Los Angeles', age: { $gt: 20 } }).explain("executionStats")

executionStats: {
    executionSuccess: true,
    nReturned: 2,
    executionTimeMillis: 1,
    totalKeysExamined: 0,
    totalDocsExamined: 6,
Drop All Indexes Except the Default _id **Index**
db.students.dropIndexes()

Aim: Data Validation and Integrity

6a. Define a schema for the 'students' collection, specifying data types and required fields. Add a custom validation rule to ensure that the email field is a valid email format.

Schema validation is most useful for an established application where you have a good sense of how to organize your data. You can use schema validation in the following scenarios:

- For a users collection, ensure that the password field is only stored as a string. This validation prevents users from saving their password as an unexpected data type, like an image.
- For a sales collection, ensure that the item field belongs to a list of items that your store sells. This validation prevents a user from accidentally misspelling an item name when entering sales data.
- For a students collection, ensure that the gpa field is always a positive number. This validation prevents errors during data entry.

db.createCollection("student",{
validator:{

```
    $jsonSchema:{                          // what type of schema
            bsonType: "object",                    //type of schema
             required:[ "name","email","gender"],  // required filed in collection
 properties:{
        name:{
             bsonType:"string",
               description:"email is the required text field"},
         email:{
            bsonType:"string",
            description:"email is the required text field"},
             gender:{
             bsonType:"string",
             description:"gender is the required text field"}
 }
 }
 }
 })
```

6b. Attempt to insert a document with invalid data (e.g., an invalid email address or missing required fields) and observe the error message. Explain the error message and how it relates to the defined schema and validation rules.

Example 1 : invalid email input

```
db.student.insertOne({
          name: "adhi",
            email: 123456, // not a sting type
            gender: "male"
})
```

Ouput:

```
MongoServerError: Document failed validation
Additional information: {
  failingDocumentId: ObjectId('67efe8dc31b9cc8469fa4216'),
  details: {
    operatorName: '$jsonSchema',
    schemaRulesNotSatisfied: [
      {
        operatorName: 'properties',
        propertiesNotSatisfied: [
          {
            propertyName: 'email',
            description: 'email is the required text field',
            details: [
              {
                operatorName: 'bsonType',
                specifiedAs: { bsonType: 'string' },
                reason: 'type did not match',
```

```
            consideredValue: 123456,
            consideredType: 'int'}}}}})
```

example 2: missing field gender

```
db.student.insertOne({
        name: "adhi",
         email: "adhirck@gmail.com "
})
Output:
MongoServerError: Document failed validation
Additional information: {
  failingDocumentId: ObjectId('67efe95531b9cc8469fa4217'),
  details: {
    operatorName: '$jsonSchema',
    schemaRulesNotSatisfied: [
      {
       operatorName: 'required',
       specifiedAs: { required: [ 'name', 'email', 'gender' ] },
       missingProperties: [ 'gender' ]
      }
    ]
  }
}
```

Example 3: Correct schema

```
db.student.insertOne({
          name: "adhi",
          email: "adhirck1@gmail.com ",
        gender: "male"
})
db.student.find()
  {
    _id: ObjectId('67efea4e31b9cc8469fa4218'),
    name: 'adhi',
    email: 'adhirck1@gmail.com ',
    gender: 'male'
  }
```

Lab 7: Aim: Creating clusters using MongoDB Atlas

7a.Create a free MongoDB Atlas cluster and connect to it using the MongoDB Shell.

What is MongoDB Atlas?

MongoDB Atlas is a fully managed **multi-cloud** database. It handles all the complexity of **deploying** and **managing** on cloud service providers like **AWS**, **Google Cloud Platform** and **Microsoft Azure**. **Atlas** is a good way to deploy, run and scale MongoDB in the cloud. Atlas offers a **free tier** (M0) which makes it an excellent choice for small projects, testing, and learning MongoDB in the cloud without incurring any cost.

What is MongoDB Shell?

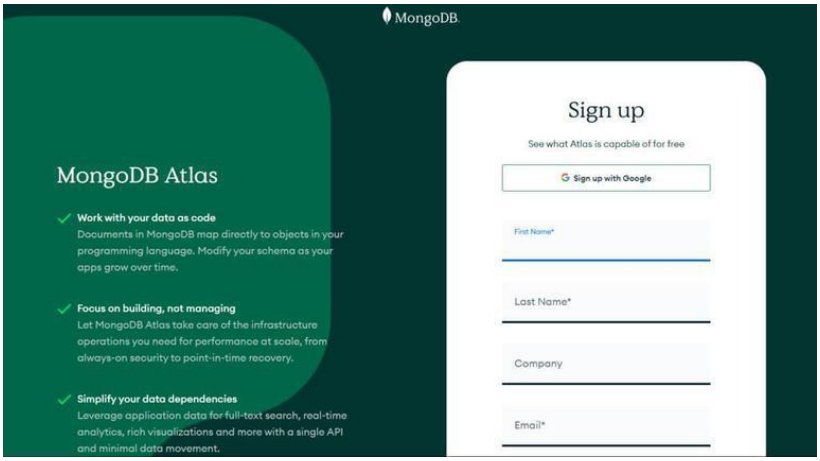
The **MongoDB Shell** (also called **mongosh**) is a JavaScript-based interface that allows you to interact with MongoDB databases from the command line. It enables **data manipulation**, querying, and performing

administrative tasks on the MongoDB database. MongoDB Shell is a powerful tool for developers working with MongoDB in local or cloud-based environments like **MongoDB Atlas**.

Step 1. Creating a Cluster in MongoDB Atlas

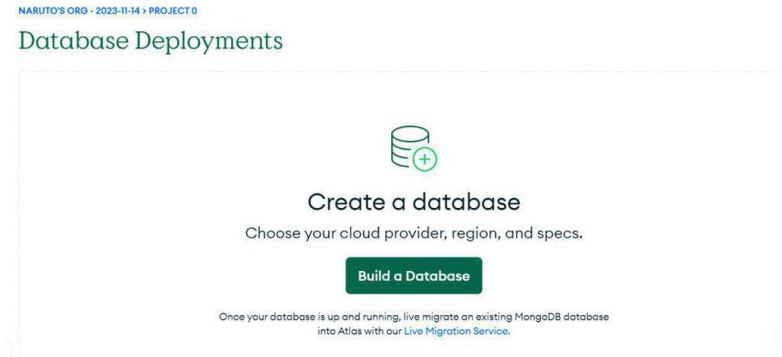
1. Sign Up for MongoDB Atlas

Go to the **MongoDB Atlas** website and sign up for an account, We can sign up using your **email** or directly with **Google**.



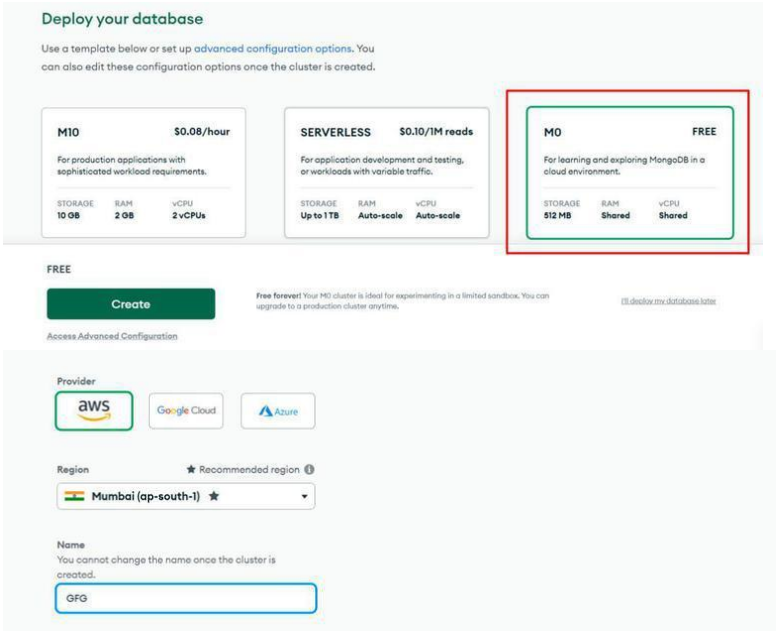
2. Build a Database

Now, go to the database option and click on the "**Build a Database**" button.



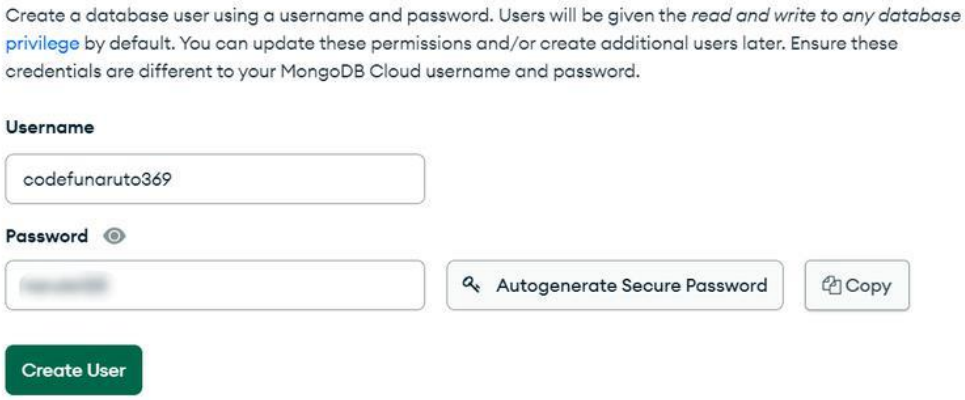
Step 2: Configure Your Cluster

- Select the **Free Tier Cluster (M0)** option.
- Choose your preferred cloud provider: **AWS**, **Google Cloud**, or **Microsoft Azure**.
- Select the **region** where you want the cluster to be hosted (e.g., North Virginia, Frankfurt, etc.).
- Rename the cluster if desired, then click **"Create"** to set up the free cluster.



Step 3: *Set Up Access*

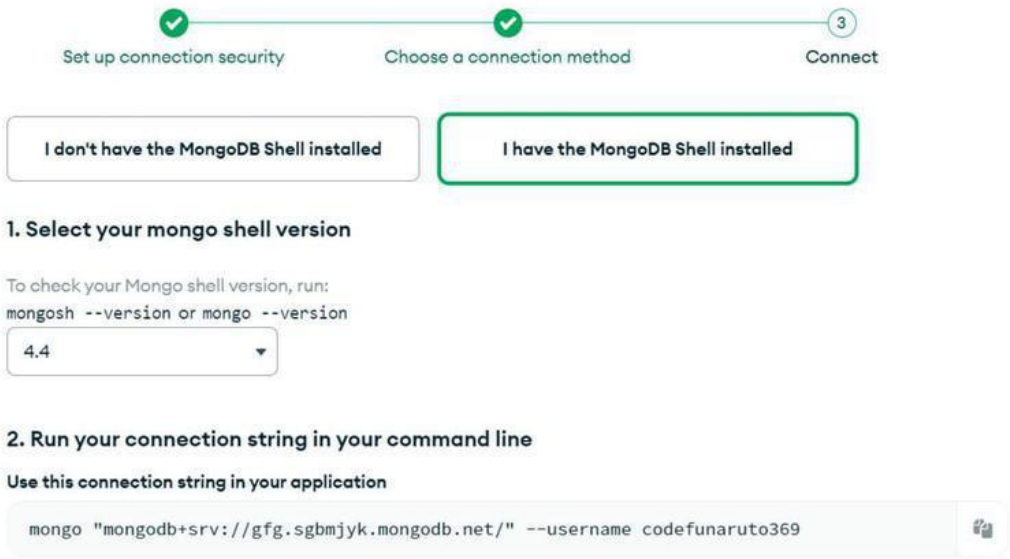
Create a **username** and **password** for the MongoDB user and configure network access settings, allowing you to connect from any IP address. These credentials will be used to connect to MongoDB Atlas.



Step 4. Establishing a Connection to MongoDB Atlas from Mongo Shell

1. Connect Using Mongo Shell

- In the MongoDB Atlas dashboard, click on the **"Connect"** button for your newly created cluster.
- Choose **"Connect with MongoDB Shell"**.
- Select the appropriate **Mongo Shell version** (Mongosh) and follow the instructions.



Step 5:Run the Connection Command

Copy the connection string and run the command in our terminal, Then we can enter the password that we have set earlier with the username and password

Step 6: Writing Queries in Mongo Shell

1. Show Databases

This command lists all the databases in your MongoDB cluster:

```
show dbs
```

2. Create and Switch to a Database

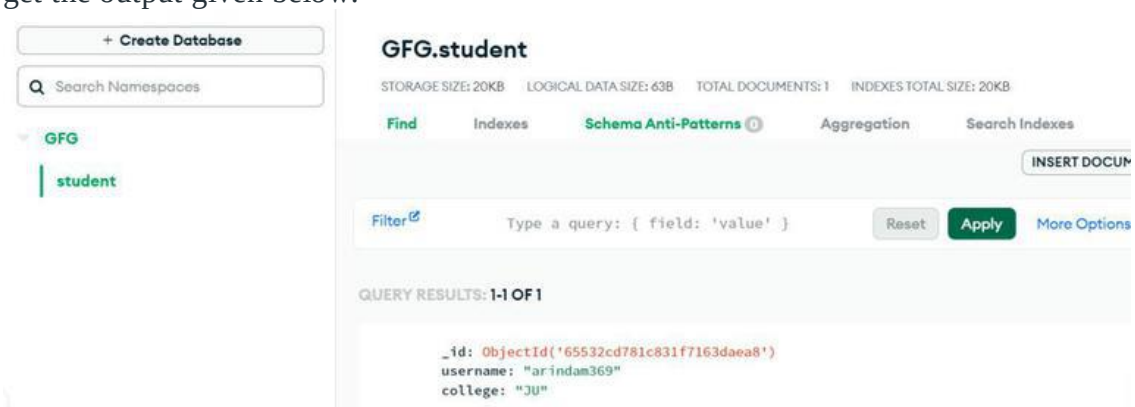
Create a new database (if it doesn't exist) and switch to it:

3. Insert a Document into a Collection

```
db. student.insertOne({"username": "arindam369",  "college": "JU"})
```

Output:

After executing the queries in the Mongo Shell, open the Collections in the **MongoDB Atlas Cluster**. We get the output given below.

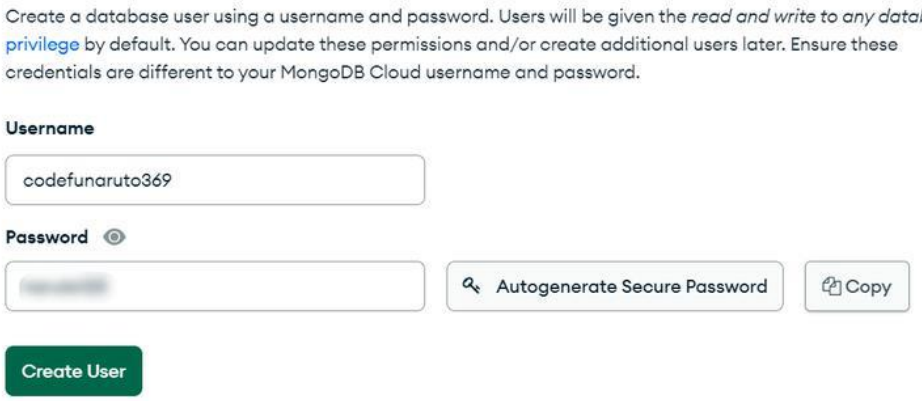


Aim: User Management and Access Control

8a. Create a new user with read-only access to the 'students' database. Try to perform write operations using this user and analyse the results. 8b. Create a role with read and write access to the 'books' collection and assign this role to a new user. Verify that the user has the expected permissions.

Step 3: *Set Up Access*

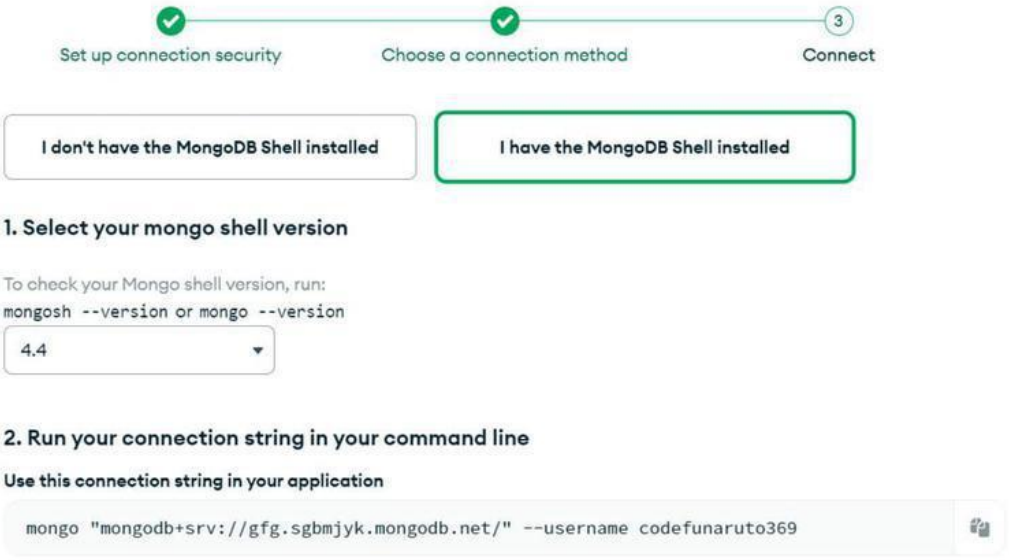
Create a **username** and **password** for the MongoDB user and configure network access settings, allowing you to connect from any IP address. These credentials will be used to connect to MongoDB Atlas.



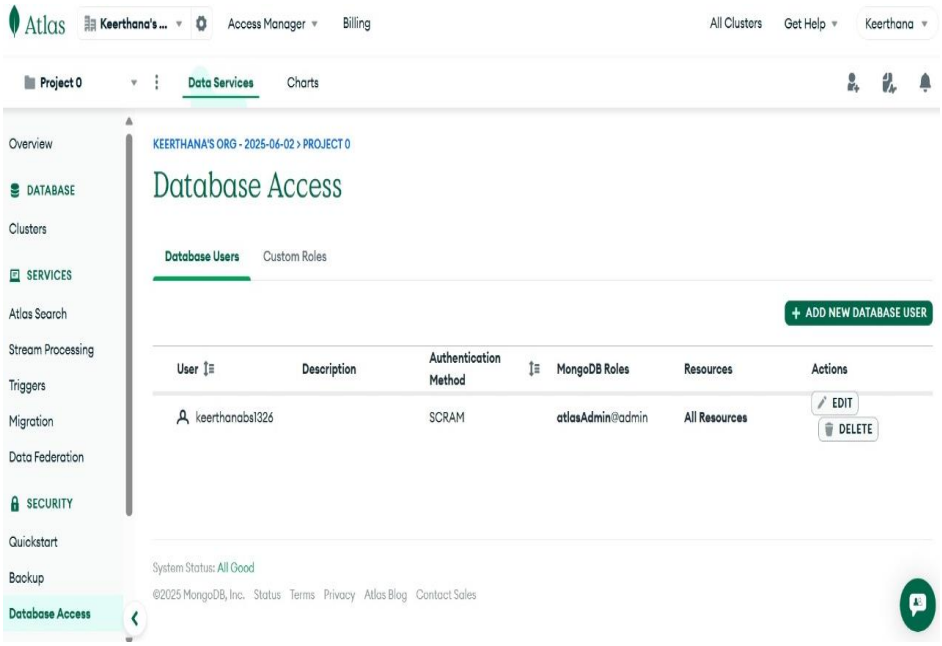
Step 4. Establishing a Connection to MongoDB Atlas from Mongo Shell

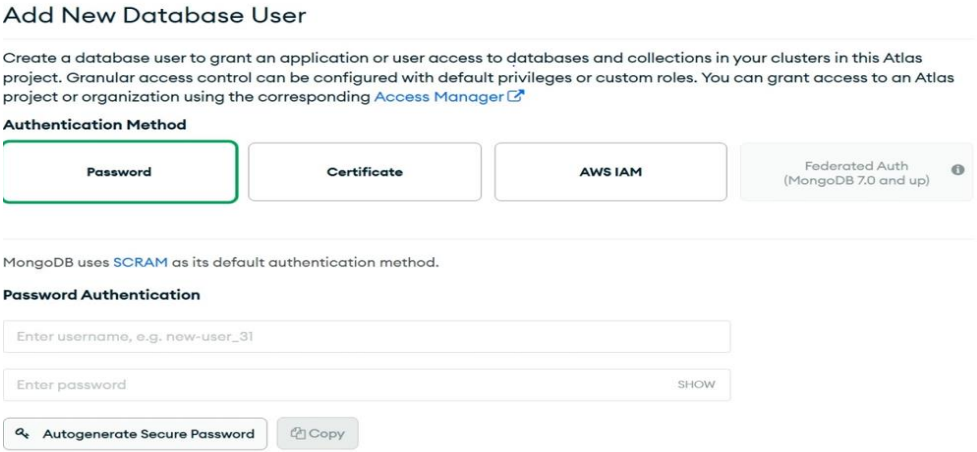
1. Connect Using Mongo Shell

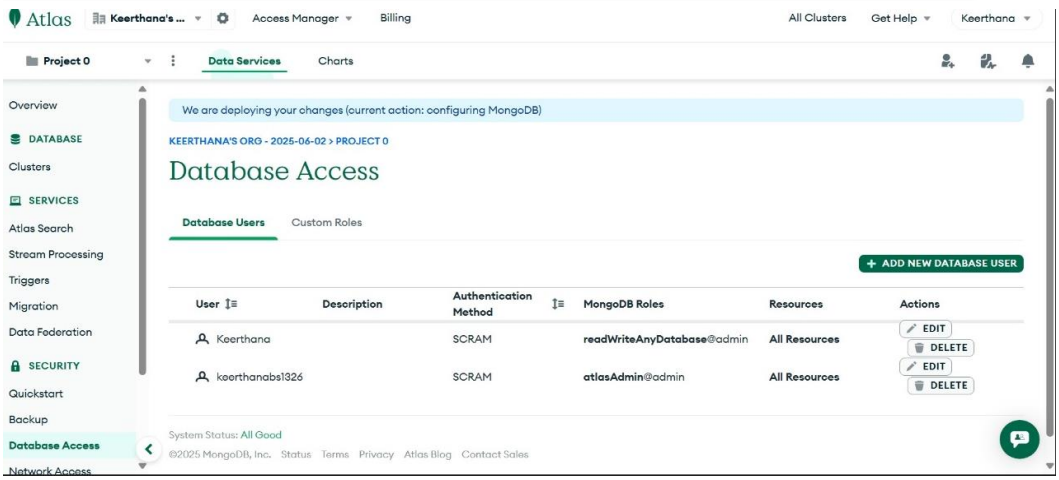
- In the MongoDB Atlas dashboard, click on the **"Connect"** button for your newly created cluster.
- Choose **"Connect with MongoDB Shell"**.
- Select the appropriate **Mongo Shell version** (Mongosh) and follow the instructions.



Step 7 : click database access then add new database user then enter username and password. And select role of the user. Read any database. Then select ok. New user is created.







1. Connect to MongoDB (as admin)

Mongosh

Use the admin database

```
use admin
```

Create a Custom Role with Read/Write on books collection

```
db.createRole({
  role: "bookRWRole",
  privileges: [
    {
      resource: { db: "library", collection: "books" },
      actions: ["find", "insert", "update", "remove"]
```

```
    }
  ],
  roles: []
})
```
Expected Output:
```
db.createUser({
  user: "bookUser",
  pwd: "bookPass123",
  roles: [{ role: "bookRWRole", db: "admin" }]
})
```
Expected Output:
```
Successfully added user: {
  "user" : "bookUser",
  "roles" : [
    {
      "role" : "bookRWRole",
      "db" : "admin"
    }
  ]
}
```
Switch to New User
```
mongosh -u bookUser -p bookPass123 --authenticationDatabase admin
```
Use the library database
```
use library
```
Try Performing Write Operation
```
db.books.insertOne({ title: "MongoDB Basics", author: "John Doe" })
```
Expected Output:
```
{
  acknowledged: true,
  insertedId: ObjectId("...")
}
```
Try Reading
```
db.books.find()
```
Expected Output:
```
[
  { _id: ObjectId("..."), title: "MongoDB Basics", author: "John Doe" }
]
```
Try Writing to Unauthorized Collection (Optional Test)
```
db.otherCollection.insertOne({ test: 1 })
```
Expected Output
```
{
  ok: 0,
  errmsg: 'not authorized on library to execute command',
  code: 13,
  codeName: 'Unauthorized'
}
```
Lab 9: Aim: MongoDB and Python

9a. Using the 'pymongo' library, connect to a local MongoDB instance. Retrieve all documents from the 'students' collection and print them to the console
```
from pymongo import MongoClient
# Connect to local MongoDB server
client = MongoClient("mongodb://localhost:27017/")
# Access the 'students' database and 'student' collection
db = client['students']
collection = db['student']
```

```
# Retrieve all documents
all_students = collection.find()
# Print each document
print("All Students:")
for student in all_students:
    print(student)
```

9.b Write a Python script to insert a new document into the 'students' collection using the pymongo library. Then, retrieve the newly inserted document and print it

```
from pymongo import MongoClient
# Connect to MongoDB (change host, port, and credentials if needed)
client = MongoClient("mongodb://localhost:27017/")
# Access the 'students' database and 'students' collection
db = client['students']
collection = db['students']
# Define the new document
new_student = {
    "name": "Alice Smith",
    "age": 20,
    "major": "Computer Science"
}
# Insert the document
insert_result = collection.insert_one(new_student)
print(f"Inserted document ID: {insert_result.inserted_id}")
# Retrieve the inserted document using the returned _id
retrieved_student = collection.find_one({"_id": insert_result.inserted_id})
# Print the retrieved document
print("Retrieved Document:")
print(retrieved_student)
```

10a. Setting up MongoDB Atlas Cluster and Configuring Security

Objective

To set up a MongoDB Atlas database cluster on a cloud provider (AWS, Azure, or GCP) and configure basic security settings for controlled access. This step is foundational for integrating with server less applications or cloud services

1. Create a MongoDB Atlas Account and Cluster

a. Sign Up / Log In

- Navigate to https://www.mongodb.com/cloud/atlas
- Create an account using your **email**, **Google**, or **GitHub**.
- After logging in, you'll be redirected to the **Atlas Dashboard**.

b. Create a New Project

- Click on **"New Project"**
- Give your project a name (e.g., StudentDBProject)
- Click **"Next"**

c. Create a Cluster

- Choose **"Build a Cluster"**
- Select **"Shared Cluster" (M0 Free Tier)** – this is sufficient for learning and experimentation.
- Name your cluster (e.g., Cluster0)
- Proceed to cloud provider selection.

2. Choose a Cloud Provider and Region

a. Choose a Provider

- Pick one of the following:
    - **AWS** (default and recommended for beginners)

- **Azure**
- **Google Cloud Platform (GCP)**

b. Choose a Region

- Choose a region closest to your location for better performance. Example:
  - **Mumbai** – if you are in India
  - **US East (N. Virginia)** – if you are in the US

c. Create Cluster

- Click **"Create Cluster"**
- Wait for a few minutes while the cluster is provisioned.

3. Configure Security Settings

A. Create a Database User

1. In the left panel, go to **"Database Access"**
2. Click **"Add New Database User"**
3. Enter:
   - **Username:** studentUser
   - **Password:** SecurePass123! (or generate securely)
4. Select **Database User Privileges**:
   - Role: Read and write to any database
5. Click **"Add User"**

B. Whitelist IP Address

1. Go to **"Network Access"**
2. Click **"Add IP Address"**
3. You have two options:
   - To **allow your current IP**, click **"Add Current IP Address"**
   - To allow access from any location temporarily (not secure), use: